

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTETC502 - DIGITAL SIGNAL PROCESSING

[Date : - 30/01/2026]

[Time : 3 Hours]

[Total Marks : 60]

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The product of two odd signals is:
 - a) Odd
 - b) Even
 - c) Both (a) and (b)
 - d) Zero
- 2 Which of the following is/are standard test signals?
 - a) Step
 - b) Impulse
 - c) Exponential
 - d) All of the above
- 3 Two-sided Z-transform is also known as:
 - a) Unilateral Z-transform
 - b) Bilateral Z-transform
 - c) Trilateral Z-transform
 - d) None of the above
- 4 IDFT of the sequence $\{1, 0, 1, 0\}$ is:
 - a) $\{1, 0, 0, 1\}$
 - b) $\{0.5, 0, 0.5, 0\}$
 - c) $\{0.5, 1, 0.5, 0\}$
 - d) None
- 5 What is the approximate transition width of main lobe of a Hamming window?
 - a) $4\pi/M$
 - b) $8\pi/M$
 - c) $12\pi/M$
 - d) $2\pi/M$

- 6 In the Bilinear Transformation mapping, which of the following are correct?
 a) All points in the LHP of s are mapped inside the unit circle in the z -plane
 b) All points in the RHP of s are mapped outside the unit circle in the z -plane
 c) Both a & b
 d) None of the mentioned
- 7 What is the lowest order of the Butterworth filter with a pass band gain $K_P = -1$ dB at $\Omega_P = 4$ rad/sec and stop band attenuation greater than or equal to 20dB at $\Omega_S = 8$ rad/sec?
 a) 4
 b) 5
 c) 6
 d) 3
- 8 With $u(n)$ as unit step function, $u(n) - u(n-1)$ is
 a) $\delta(n)$
 b) $\delta(n) + \delta(n-1)$
 c) $\delta(n) - \delta(n-1)$
 d) $\delta(n) + \delta(n+1)$
- 9 Which of the following is true regarding the number of computations required to compute an N -point DFT?
 a) N^2 complex multiplications and $N(N-1)$ complex additions
 b) N^2 complex additions and $N(N-1)$ complex multiplications
 c) N^2 complex multiplications and $N(N+1)$ complex additions
 d) N^2 complex additions and $N(N+1)$ complex multiplications
- 10 The z -transform of the system $h(n) = 3^n u(n)$ is:
 a) $3z / z - 3$
 b) $z / z + 3$
 c) $z / z + 3$
 d) $z / z - 3$
- 11 Which of the following statement is/are correct about linear convolution?
 a) The Input and output sequence is Aperiodic
 b) It requires zero padding
 c) The length of the input and output sequence is the same.
 d) The length of output sequence is greater than the input sequence
- 12 For $x(n) = -u(-n-1)$; then ROC is
 a) $|z| > 1$
 b) $|z| < 1$
 c) $|z| > 0$
 d) $|z| < 0$

Q2. Solve the following.

- A) Explain the representation of discrete time signal in detail. 6
 B) Find impulse response of the system given by $y(n) - 5y(n-1) + 6y(n-2) = x(n)$. 6

Q3. Solve the following.

- A) Find IDFT of the sequence given by $X(K) = \{2, 1-j, 0, 1+j\}$. 6
 B) Define sampling theorem and explain aliasing effect. 6

Q4. Solve Any Two of the following.

- A) Find circular convolution using concentric circle method $x(n) = \{2, 2, 1, 1\}$ and $h(n) = \{1, 1, 2\}$. 6

~~B)~~ Find four point DFT of sequence $x(n) = \{1, 1, 2, 2\}$ using DIT FFT algorithm 6

~~C)~~ State the properties of ROC of Z transform. 6

Q5. Solve Any Two of the following.

~~A)~~ Using BLT method obtain $H(z)$ if $H(s) = 1/(s+1)^2$ 6

B) The desired frequency response of a lowpass filter is 6

$$H_d(e^{j\omega}) = e^{-j\omega} \text{ for } -3\pi/4 \leq \omega \leq 3\pi/4 \text{ and}$$

$$H_d(e^{j\omega}) = 0 \text{ for } 3\pi/4 \leq \omega \leq \pi$$

Design filter using windowing technique use Hamming window $N = 5$

~~C)~~ Find the Z transform and ROC of the signal $x(n) = -a^n u(-n-1)$ 6

Q6. Solve Any Two of the following.

done ~~A)~~ Derive the equation for the order of the Low pass butterworth analog filter R 6

done ~~B)~~ Discuss the properties of DTFT R 6

C) What is the need of multirate signal processing and give some examples of multirate digital systems and explain any one R 6

*** End ***